

ABOUT THE TEST FoundationOne® Heme is a comprehensive genomic profiling test designed to identify genomic alterations within hundreds of cancer-related genes in hematologic malignancies and sarcomas.

PATIENT

DISEASE Bone marrow leukemia lymphocytic chronic (CLL)
NAME
DATE OF BIRTH
SEX
MEDICAL RECORD #

PHYSICIAN

ORDERING PHYSICIAN
MEDICAL FACILITY
ADDITIONAL RECIPIENT
MEDICAL FACILITY ID
PATHOLOGIST

SPECIMEN

SPECIMEN SITE
SPECIMEN ID
SPECIMEN TYPE
DATE OF COLLECTION
SPECIMEN RECEIVED

Biomarker Findings

Microsatellite status - MS-Stable

Tumor Mutational Burden - TMB-Low (1 Muts/Mb)

Genomic Findings

For a complete list of the genes assayed, please refer to the Appendix.

DNMT3A F732del

TET2 splice site 3500+2T>C - subclonal†

CHEK2 splice site 444+1G>A

XPO1 E571K

† See About the Test in appendix for details.

2 Therapies with Clinical Benefit

2 Clinical Trials

0 Therapies with Lack of Response

BIOMARKER FINDINGS

Microsatellite status - MS-Stable

Tumor Mutational Burden - TMB-Low (1 Muts/Mb)

GENOMIC FINDINGS

DNMT3A - F732del

2 Trials see p. 7

TET2 - splice site 3500+2T>C - subclonal

2 Trials see p. 8

ACTIONABILITY

No therapies or clinical trials. see Biomarker Findings section

No therapies or clinical trials. see Biomarker Findings section

THERAPIES WITH CLINICAL BENEFIT (IN PATIENT'S TUMOR TYPE)

none

THERAPIES WITH CLINICAL BENEFIT (IN OTHER TUMOR TYPE)

Azacitidine

Decitabine

none

Azacitidine

Decitabine

GENOMIC FINDINGS WITH NO REPORTABLE THERAPEUTIC OR CLINICAL TRIALS OPTIONS

For more information regarding biological and clinical significance, including prognostic, diagnostic, germline, and potential chemosensitivity implications, see the Genomic Findings section.

CHEK2 - splice site 444+1G>A

p. 5

XPO1 - E571K

p. 5

NOTE Genomic alterations detected may be associated with activity of certain FDA-approved drugs; however, the agents listed in this report may have varied clinical evidence in the patient's tumor type. Neither the therapeutic agents nor the trials identified are ranked in order of potential or predicted efficacy for this patient, nor are they ranked in order of level of evidence for this patient's tumor type.

TRF#

BIOMARKER FINDINGS

BIOMARKER

Microsatellite status

CATEGORY

MS-Stable

POTENTIAL TREATMENT STRATEGIES

On the basis of clinical evidence, MSS tumors are significantly less likely than MSI-H tumors to respond to anti-PD-1 immune checkpoint inhibitors¹⁻³, including approved therapies nivolumab and pembrolizumab⁴. In a retrospective analysis of 361 patients with solid tumors treated with pembrolizumab, 3% were MSI-H and experienced a significantly higher ORR compared with non-MSI-H cases (70% vs. 12%, $p=0.001$)⁵.

FREQUENCY & PROGNOSIS

In a computational analysis of paired tumor and normal cases in the TCGA dataset, 1 of 338

chronic lymphocytic leukemia (CLL) samples (0.3%) was MSI-H⁶. In a large study of 940 patients with B-cell CLL, fewer than 1% of cases demonstrated MSI, and none of the cases were MSI-H⁷. In smaller studies of CLL, MSI at any level has been observed in 0% (0/39) to 52% (10/19) of cases⁸⁻¹⁴, and MSI-H has been observed in 5.3% (1/19) to 21% (4/19) of cases^{9-11,13} or reported as absent^{7,12,14-15}. High MSI (MSI-H) is generally rare in hematologic malignancies compared with solid tumors. Moreover, reports of MSI in hematologic malignancies in the literature are conflicting and varied due to substantial heterogeneity, lack of consensus on the markers and methods used for MSI assessment, small sample size in most studies, and possible elimination of MSI-positive cells in the bloodstream by immunosurveillance¹⁶. In studies of Richter's transformation of CLL, some publications have reported an association of MSI-H with transformation⁹⁻¹⁰, whereas others have not¹¹⁻¹².

FINDING SUMMARY

Microsatellite instability (MSI) is a condition of genetic hypermutability that generates excessive amounts of short insertion/deletion mutations in the genome; it generally occurs at microsatellite DNA sequences and is caused by a deficiency in DNA mismatch repair (MMR) in the tumor¹⁷. Defective MMR and consequent MSI occur as a result of genetic or epigenetic inactivation of one of the MMR pathway proteins, primarily MLH1, MSH2, MSH6, or PMS2¹⁷⁻¹⁹. This sample is microsatellite-stable (MSS), equivalent to the clinical definition of an MSS tumor: one with mutations in none of the tested microsatellite markers²⁰⁻²². MSS status indicates MMR proficiency and typically correlates with intact expression of all MMR family proteins^{17,19,21-22}.

TRF#

BIOMARKER FINDINGS

BIOMARKER

Tumor Mutational Burden

CATEGORY

TMB-Low (1 Muts/Mb)

POTENTIAL TREATMENT STRATEGIES

On the basis of emerging clinical evidence, increased TMB may be associated with greater sensitivity to immunotherapeutic agents, including anti-CTLA-4²³, anti-PD-L1²⁴⁻²⁷, and anti-PD-1 therapies^{4,28-29}; FDA-approved agents include ipilimumab, atezolizumab, avelumab, durvalumab, pembrolizumab, and nivolumab. In multiple solid tumor types, higher mutational burden has corresponded with response and improved prognosis. Pembrolizumab improved progression-free survival (14.5 vs. 3.4-3.7 months) for patients with non-small cell lung cancer (NSCLC) and higher mutational load (greater than 200 nonsynonymous mutations; hazard ratio = 0.19)²⁸. In studies of patients with either NSCLC or colorectal cancer (CRC), patients whose tumors harbored elevated mutational burden reported higher overall response rates to pembrolizumab^{4,28-29}. Anti-PD-1 therapies have achieved clinical benefit for certain patients with high mutational burden, including 3 patients with endometrial adenocarcinoma who reported sustained partial responses (PRs) following treatment

with pembrolizumab³⁰ or nivolumab³¹, a patient with hypermutant glioblastoma who obtained clinical benefit from pembrolizumab³², 2 pediatric patients with biallelic mismatch repair deficiency-associated ultrahypermutant glioblastoma who experienced clinically and radiologically significant responses to nivolumab³³, and 2 patients with microsatellite-stable rectal cancers, 1 who achieved an ongoing PR to pembrolizumab and the other an ongoing complete response to nivolumab³⁴. For patients with melanoma, mutational load was associated with long-term clinical benefit from ipilimumab^{23,35} and anti-PD-1/anti-PD-L1 treatments²⁵. For patients with metastatic urothelial carcinoma (mUC), those who responded to atezolizumab treatment had a significantly increased mutational load (12.4 mutations [mut] per megabase [Mb]) compared to nonresponders (6.4 muts/Mb)²⁴, and mutational load of 16 muts/Mb or higher was associated with significantly longer overall survival²⁶. In a retrospective analysis of 17 solid tumor types (comprised of 47% NSCLC, 40% mUC, and 13% encompassing 15 other solid tumors), a TMB of ≥16 muts/Mb associated with an objective response rate to atezolizumab of 30% vs. 14% for chemotherapy alone³⁶.

FREQUENCY & PROGNOSIS

Chronic lymphocytic leukemia (CLL) harbors a median TMB of 1.7 mutations per megabase

(mut/Mb), and none of the cases analyzed in one genomic study had high TMB (>20 muts/Mb)³⁷. Reports of high TMB are generally rare in leukemia³⁷. In a study of 92 patients with various hematologic malignancies, elevated TMB levels (>10 muts/Mb) were not detected in AML (0/5) or ALL (0/1) cases analyzed³⁸. The prognostic significance of TMB in CLL has not been extensively studied (PubMed, Oct 2018).

FINDING SUMMARY

Tumor mutational burden (TMB, also known as mutation load) is a measure of the number of somatic protein-coding base substitution and insertion/deletion mutations occurring in a tumor specimen. TMB is affected by a variety of causes, including exposure to mutagens such as ultraviolet light in melanoma³⁹⁻⁴⁰ and cigarette smoke in lung cancer^{28,41}, mutations in the proofreading domains of DNA polymerases encoded by the POLE and POLD1 genes⁴²⁻⁴⁶, and microsatellite instability (MSI)^{42,45-46}. This sample harbors a low TMB. Compared to patients with tumors harboring higher TMB levels, patients with tumors harboring low TMB levels have experienced lower rates of clinical benefit from treatment with immune checkpoint inhibitors, including anti-CTLA-4 therapy in melanoma²³, anti-PD-L1 therapy in urothelial carcinoma²⁴, and anti-PD-1 therapy in non-small cell lung cancer and colorectal cancer^{4,28}.

TRF#

GENOMIC FINDINGS
GENE
DNMT3A
ALTERATION
F732del
POTENTIAL TREATMENT STRATEGIES

DNA methyltransferase (DNMT) inhibitors, including azacitidine and decitabine, have been used with some success in the treatment of MDS and AML⁴⁷, and patients with AML or MDS harboring DNMT3A mutations have been reported to benefit from DNMT inhibitors⁴⁸⁻⁵¹, including patients with R882 mutations, although the data are not conclusive. Mutations in DNMT3A have been associated with improved rate of survival for patients treated with high-dose daunorubicin, compared with standard-dose daunorubicin⁵². Other studies have shown that when treated with idarubicin, patients with exon 23 mutation in DNMT3A (including R882) had better median disease-free survival and overall survival compared with patients lacking exon 23 mutations, although a similar response was not observed for patients treated with daunorubicin⁵³. Mutations in DNMT3A have

been associated with improved rate of survival for patients treated with high-dose daunorubicin, compared with standard-dose daunorubicin⁵². Other studies have shown that when treated with idarubicin, patients with exon 23 mutation in DNMT3A (including R882) had better median disease-free survival and overall survival compared with patients lacking exon 23 mutations, although a similar response was not observed for patients treated with daunorubicin⁵³. It is not known whether these therapeutic approaches would be relevant in the context of alterations that have not been fully characterized, as seen here.

FREQUENCY & PROGNOSIS

In the COSMIC database, DNMT3A mutation was observed in <1% of chronic lymphocytic leukemia (CLL) cases (Jan 2019). However, DNMT3A has been shown to be inactivated by TCL1 that is frequently overexpressed in CLL⁵⁴. DNMT3A transcript level was reduced in high/intermediate risk CLL samples compared to low risk CLL⁵⁵. Reduced DNMT3A expression was also noted in samples from patient with CLL at PD compared to at SD, and reduction in DNMT3A

levels correlated with shorter time to treatment in univariate ($P=0.05$), but not multivariate analysis⁵⁶. Mutations in genes including ASXL1, CBL, DNMT3A, MYD88, JAK2, TET2, SF3B1, and U2AF1 have been associated with clonal hematopoiesis of indeterminate potential (CHIP), which is age-related and associated with increased risk of hematologic cancers⁵⁷⁻⁶¹.

FINDING SUMMARY

The DNMT3A gene encodes the protein DNA methyltransferase 3A, an enzyme that is involved in the methylation of newly synthesized DNA, a function critical for gene regulation⁶²⁻⁶³. The role of DNMT3A in cancer is uncertain, as some reports describe increased expression and contribution to tumor growth, whereas others propose a role for DNMT3A as a tumor suppressor⁶⁴⁻⁶⁹. Although alterations such as seen here have not been fully characterized and are of unknown functional significance, similar alterations have been previously reported in the context of cancer, which may indicate biological relevance.

GENE
TET2
ALTERATION
splice site 3500+2T>C - subclonal
POTENTIAL TREATMENT STRATEGIES

TET2 loss or inactivating mutations may lead to increased DNA methylation and may predict sensitivity to DNA methyltransferase (DNMT) inhibitors such as the FDA-approved therapies azacitidine and decitabine. TET2 mutation status in myelodysplastic syndrome (MDS) was significantly associated with better response rates to the DNMT inhibitors azacitidine and/

or decitabine^{48,70-71}. In other clinical studies, patients with TET2-mutated angioimmunoblastic T-cell lymphoma (AITL) were reported to achieve complete responses to azacitidine⁷²⁻⁷⁴.

FREQUENCY & PROGNOSIS

TET2 mutations have been reported in <1% chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL) cases in the COSMIC database (Oct 2018), and were not detected in other studies of CLL (0/48 and 0/75)⁷⁵⁻⁷⁶. TET2 overexpression was detected in a study of B-cells from 48 patients with CLL; however, no TET2 mutations were detected⁷⁵. High TET2 expression is associated with longer median treatment-free survival in

patients with CLL⁷⁷. Mutations in genes including ASXL1, CBL, DNMT3A, MYD88, JAK2, TET2, SF3B1, and U2AF1 have been associated with clonal hematopoiesis of indeterminate potential (CHIP), which is age-related and associated with increased risk of hematologic cancers⁵⁷⁻⁶¹.

FINDING SUMMARY

TET2 encodes a tumor suppressor involved in reversing DNA methylation marks, a process critical for proper gene regulation⁷⁸⁻⁷⁹. TET2 alterations that impact critical residues or result in the disruption or loss of the catalytic domain (amino acids 1129-1936), such as seen here, are predicted to impair the tumor suppressor activity of TET2⁸⁰⁻⁸⁴.

TRF#

GENOMIC FINDINGS

GENE

CHEK2

ALTERATION

splice site 444+1G>A

POTENTIAL TREATMENT STRATEGIES

There are no FDA-approved or investigational therapies targeting CHEK2 loss or inactivating alterations. One study of patients with breast cancer reported that carriers of the CHEK2 H371Y mutation have a higher likelihood of response to neoadjuvant chemotherapy⁸⁵, whereas another study found that CHEK2 mutation carriers have a lower frequency of objective clinical responses to neoadjuvant therapy⁸⁶. A third study reported that the CHEK2 1100delC mutation is not associated with differential efficacy of chemotherapy and endocrine therapy in patients with metastatic breast cancer⁸⁷.

FREQUENCY & PROGNOSIS

CHEK2 mutations have been reported in 7% of glioblastoma (GBM) samples and in carcinomas of the ovary (2%), endometrium (1.9%), and large intestine (1.7%), as well as in a variety of solid and hematologic cancer types at low frequency (COSMIC, 2018). In breast cancer, certain CHEK2 mutations are associated with higher grade and larger tumors as well as bilateral disease⁸⁸. A study reported that a polymorphism in CHEK2 was associated with worse survival of patients with GBM, but this association lost significance after adjusting for other prognostic factors⁸⁹⁻⁹⁰. Another study in prostate cancer reported that CHEK2 expression is decreased in higher grade tumors and that CHEK2 is a tumor suppressor that decreases the growth of prostate cancer cells and regulates androgen receptor signaling⁹¹.

FINDING SUMMARY

CHEK2 encodes the protein checkpoint kinase 2, a serine/threonine kinase that plays an important role in the DNA-damage response; it is a putative tumor suppressor⁹²⁻⁹⁵. Germline mutations in CHEK2 have been associated with uterine serous carcinoma cases and with an increased risk of breast, colorectal, and thyroid cancer, as well as non-Hodgkin lymphoma⁹⁶⁻¹⁰³. CHEK2 alterations that disrupt or remove the SQ/TQ cluster domain (SCD; amino acids 19-69), forkhead-associated domain (FHA; amino acids 115-175), and/or the kinase domain (amino acids 220-486) are predicted to be inactivating¹⁰⁴⁻¹⁰⁶. Some CHEK2 alterations, including T367fs*15 (also known as 1100delC) and S428F, have been reported as germline variants in low penetrance susceptibility to certain cancers, including breast and ovarian¹⁰⁷⁻¹¹¹.

GENE

XPO1

ALTERATION

E571K

POTENTIAL TREATMENT STRATEGIES

XPO1 activation may predict sensitivity to XPO1 inhibitors¹¹², such as the selective inhibitors of nuclear export (SINEs)¹¹³ KPT-330¹¹⁴ and KPT-8602¹¹⁵⁻¹¹⁶, which are in clinical trials in solid tumors and hematological cancers. In a Phase 1 trial of KPT-330 for unselected advanced solid tumors, one of 157 evaluable patients achieved a complete response, six achieved partial responses (4%), and 67 (43%) achieved stable disease (SD), which was durable for ≥4 months for 27 patients (17%)¹¹⁷. A patient with accelerated-phase chronic myelogenous leukemia (CML) refractory to available tyrosine kinase inhibitors demonstrated short-term anti-leukemic activity from KPT-330¹¹⁸.

Inhibitors of XPO1 have shown efficacy in mouse models and cultured cells of CML, T-cell acute lymphoblastic leukemia, acute myeloid leukemia, mantle cell lymphoma, multiple myeloma, and CLL^{114,116,118-123}, as well as in GIST, a subset of sarcomas¹²⁴, and ESCCs¹²⁵.

FREQUENCY & PROGNOSIS

XPO1 mutations have been reported at highest frequency in CLL (2-15%) and endometrial tumors (3-5%) and at lower incidences in other tumor types (COSMIC, 2018)^{42,126-129}. One study reported XPO1 mutation at high frequency in cases of Hodgkin lymphoma (26%, 5/19) and primary mediastinal B-cell lymphoma (24%, 28/117)¹³⁰. In CLL, XPO1 correlates with unmutated IGHV status, presence of NOTCH1 mutations, increased white blood cell count, and decreased progression-free survival after certain chemoimmunotherapy regimens^{126-127,129,131}. Increased levels of nuclear exportin-1 have

been reported to correlate with clinical stage in human cholangiocarcinoma samples, and reduced exportin-1 expression suppressed tumor growth in a xenograft model of the disease¹³².

FINDING SUMMARY

XPO1 encodes exportin-1, also known as CRM1, a nuclear pore protein that controls nuclear export and localization of multiple cell cycle- and proliferation-associated proteins¹³³. Exportin-1 may also be required for efficient centrosome maintenance by BRCA1¹³⁴. XPO1 residue E571 is a mutational hotspot in chronic lymphocytic leukemia (CLL) and other tumor types^{126-127,129-131,135}, although mutations at this site have not been functionally characterized. XPO1 D624G has been reported in the context of esophageal squamous cell carcinoma (ESCC) and has been suggested to lead to gain of function¹²⁵.

TRF#

THERAPIES WITH CLINICAL BENEFIT

IN OTHER TUMOR TYPE

Azacitidine

Assay findings association

DNMT3A
F732del

TET2
splice site 3500+2T>C - subclonal

AREAS OF THERAPEUTIC USE

Azacitidine is an injectable nucleoside analog that acts as a DNA methyltransferase inhibitor. It is FDA approved for the treatment of patients with myelodysplastic syndrome (MDS). It is also approved in combination with Venetoclax for the treatment of patients 75 years of age or older with newly diagnosed acute myeloid leukemia (AML) or comorbidities that preclude use of intensive induction chemotherapy.

GENE ASSOCIATION

DNMT3A neomorphic or gain-of-function mutations may result in increased DNA methylation, and may predict sensitivity to DNA methyltransferase inhibitors. Some patients with acute myeloid leukemia (AML) or MDS and DNMT3A mutations have been reported to benefit from DNMT inhibitors^{48,51,136}. On the basis of clinical studies in angioimmunoblastic T-cell lymphoma (AITL)⁷²⁻⁷⁴ and MDS^{48,70-71}, inactivating mutations in TET2 may predict

sensitivity to DNA methyltransferase (DNMT) inhibitors. It is not known whether this therapeutic approach would be relevant in the context of alterations that have not been fully characterized, as seen here.

SUPPORTING DATA

A Phase 1 trial of azacitidine followed by standard chemotherapy for patients with newly diagnosed diffuse large B-cell lymphoma (DLBCL) reported complete responses for 11 of 12 patients¹³⁷. In a study of azacitidine in combination with vorinostat and chemotherapy for patients with refractory or poor-risk relapsed lymphoma, patients with DLBCL achieved an overall response rate (ORR) of 78% and a complete response (CR) rate of 55%, patients with Hodgkin lymphoma had a CR rate of 88% (7/8), and 2/2 patients with T-cell non-Hodgkin lymphoma experienced a CR; the majority of the CRs were durable¹³⁸.

Decitabine

Assay findings association

DNMT3A
F732del

TET2
splice site 3500+2T>C - subclonal

AREAS OF THERAPEUTIC USE

Decitabine is an injectable nucleoside analog that acts as a DNA methyltransferase inhibitor. It has been approved by the FDA for the treatment of patients with myelodysplastic syndrome (MDS). It is also approved in combination with Venetoclax for the treatment of patients 75 years of age or older with newly diagnosed acute myeloid leukemia (AML) or comorbidities that preclude use of intensive induction chemotherapy.

GENE ASSOCIATION

DNMT3A neomorphic or gain-of-function mutations may result in increased DNA methylation, and may predict sensitivity to DNA methyltransferase inhibitors. Some patients with acute myeloid leukemia (AML) or MDS and DNMT3A mutations have been reported to benefit from DNMT inhibitors^{48,51,136}. On the basis of clinical studies in angioimmunoblastic T-cell lymphoma (AITL)⁷²⁻⁷⁴ and MDS^{48,70-71}, inactivating mutations in TET2 may predict

sensitivity to DNA methyltransferase (DNMT) inhibitors. It is not known whether this therapeutic approach would be relevant in the context of alterations that have not been fully characterized, as seen here.

SUPPORTING DATA

A Phase 1 study of low-dose decitabine in 4 patients with non-Hodgkin lymphoma and 16 patients with chronic lymphocytic leukemia reported dose-limiting myelosuppression and no significant changes in genome-wide DNA methylation; 8 patients in this study experienced stable disease¹³⁹. A Phase 1 trial of decitabine in combination with vorinostat in patients with advanced solid tumors and non-Hodgkin lymphomas has reported stable disease in 11/38 (29%) of patients¹⁴⁰. A preclinical study reported that decitabine could synergize with a histone deacetylase inhibitor to reduce the viability of mantle cell lymphoma cells¹⁴¹.

NOTE Genomic alterations detected may be associated with activity of certain FDA approved drugs, however the agents listed in this report may have little or no evidence in the patient's tumor type.

TRF#

CLINICAL TRIALS

IMPORTANT Clinical trials are ordered by gene and prioritized by: age range inclusion criteria for pediatric patients, proximity to ordering medical facility, later trial phase, and verification of trial information within the last two months. While every effort is made to ensure the accuracy of the information contained below, the information available in the public domain is continually

updated and should be investigated by the physician or research staff. This is not a comprehensive list of all available clinical trials. Foundation Medicine displays a subset of trial options and ranks them in this order of descending priority: Qualification for pediatric trial → Geographical proximity → Later trial phase. Clinical trials listed here may have additional enrollment criteria that

may require medical screening to determine final eligibility. For additional information about listed clinical trials or to conduct a search for additional trials, please see clinicaltrials.gov. Or visit <https://www.foundationmedicine.com/genomic-testing#support-services>.

GENE
DNMT3A

ALTERATION
F732del

RATIONALE
DNA methyltransferase (DNMT) inhibitors may offer therapeutic strategies for patients with certain mutations in DNMT3A. It is not known

whether these therapeutic approaches would be relevant in the context of alterations that have not been fully characterized, as seen here.

NCT01998035

Romidepsin Plus Oral 5-Azacitidine in Relapsed/Refractory Lymphoid Malignancies

PHASE 1/2

TARGETS
HDAC, DNMT

LOCATIONS: New York

NCT02494258

A Study to Evaluate Long-term Safety in Subjects With Solid Tumors and Hematological Disorders.

PHASE 2

TARGETS
DNMT

LOCATIONS: Florida, Maryland, Texas

TRF#

CLINICAL TRIALS

GENE

TET2

ALTERATION

splice site 3500+2T>C - subclonal

RATIONALE

 One strategy under investigation to address
 mutation or loss of TET2 in human cancer

 involves DNA methyltransferase (DNMT)
 inhibitors.

NCT01998035
PHASE 1/2

Romidepsin Plus Oral 5-Azacitidine in Relapsed/Refractory Lymphoid Malignancies

TARGETS

HDAC, DNMT

LOCATIONS: New York

NCT02494258
PHASE 2

A Study to Evaluate Long-term Safety in Subjects With Solid Tumors and Hematological Disorders.

TARGETS

DNMT

LOCATIONS: Florida, Maryland, Texas

SAMPLE

TRF#

APPENDIX
Variants of Unknown Significance

NOTE One or more variants of unknown significance (VUS) were detected in this patient's tumor. These variants may not have been adequately characterized in the scientific literature at the time this report was issued, and/or the genomic context of these alterations makes their significance unclear. We choose to include them here in the event that they become clinically meaningful in the future.

ATM
V613L

AXL
M521T

FANCC
C10Y

FOXO3
G104R

MED12
Q2119_Q2120insHQQQ

MYO18A
R587W

PDCD1LG2 (PD-L2)
V160I

POT1
I22V

SAMPLE

TRF#

APPENDIX
Genes Assayed in FoundationOne®Heme

FoundationOne Heme is designed to include genes known to be somatically altered in human hematologic malignancies and sarcomas that are validated targets for therapy, either approved or in clinical trials, and/or that are unambiguous drivers of oncogenesis based on current knowledge. The current assay utilizes DNA sequencing to interrogate 406 genes as well as selected introns of 31 genes involved in rearrangements, in addition to RNA sequencing of 265 genes. The assay will be updated periodically to reflect new knowledge about cancer biology.

HEMATOLOGICAL MALIGNANCY DNA GENE LIST: ENTIRE CODING SEQUENCE FOR THE DETECTION OF BASE SUBSTITUTIONS, INSERTION/DELETIONS, AND COPY NUMBER ALTERATIONS

ABL1	ACTB	AKT1	AKT2	AKT3	ALK	AMER1 (FAM123B or WTX)	APC
APH1A	AR	ARAF	ARFRP1	ARHGAP26 (GRAF)		ARID1A	ARID2
ASXL1	ATM	ATR	ATRX	AURKA	AURKB	AXIN1	AXL
BAP1	BARD1	BCL10	BCL11B	BCL2	BCL2L2	BCL6	BCL7A
BCORL1	BIRC3	BLM	BRAF	BRCA1	BRCA2	BRD4	BRIP1
BTG2	BTK	BTLA	C11orf30 (EMSY)	CAD	CALR*	CARD11	CBFB
CCND1	CCND2	CCND3	CCNE1	CCT6B	CD22	CD274 (PD-L1)	CD36
CD70	CD79A	CD79B	CDC73	CDH1	CDK12	CDK4	CDK6
CDKN1B	CDKN2A	CDKN2B	CDKN2C	CEBPA	CHD2	CHEK1	CHEK2
CIITA	CKS1B	CPS1	CREBBP	CRKL	CRLF2	CSF1R	CSF3R
CTNNA1	CTNNB1	CUX1	CXCR4	DAXX	DDR2	DDX3X	DNM2
DOT1L	DTX1	DUSP2	DUSP9	EBF1	ECT2L	EED	EGFR
EP300	EPHA3	EPHA5	EPHA7	EPHB1	ERBB2	ERBB3	ERBB4
ESR1	ETS1	ETV6	EXOSC6	EZH2	FAF1	FAM46C	FANCA
FANCD2	FANCE	FANCF	FANCG	FANCL	FAS (TNFRSF6)	FBXO11	FBXO31
FGF10	FGF14	FGF19	FGF23	FGF3	FGF4	FGF6	FGFR1
FGFR3	FGFR4	FHIT	FLCN	FLT1	FLT3	FLT4	FLYWCH1
FOXO1	FOXO3	FOXP1	FRS2	GADD45B	GATA1	GATA2	GATA3
GNA11	GNA12	GNA13	GNAQ	GNAS	GPR124	GRIN2A	GSK3B
HDAC1	HDAC4	HDAC7	HGF	HIST1H1C	HIST1H1D	HIST1H1E	HIST1H2AC
HIST1H2AL	HIST1H2AM	HIST1H2BC	HIST1H2BJ	HIST1H2BK	HIST1H2BO	HIST1H3B	HNF1A
HSP90AA1	ICK	ID3	IDH1	IDH2	IGF1R	IKBKE	IKZF1
IKZF3	IL7R	INHBA	INPP4B	INPP5D (SHIP)	IRF1	IRF4	IRF8
JAK1	JAK2	JAK3	JARID2	JUN	KAT6A (MYST3)	KDM2B	KDM4C
KDM5C	KDM6A	KDR	KEAP1	KIT	KLHL6	KMT2A (MLL)	KMT2C (MLL3)
KRAS	LEF1	LRP1B	LRRK2	MAF	MAFB	MAGED1	MALT1
MAP2K2	MAP2K4	MAP3K1	MAP3K14	MAP3K6	MAP3K7	MAPK1	MCL1
MDM4	MED12	MEF2B	MEF2C	MEN1	MET	MIB1	MITF
MLH1	MPL	MRE11A	MSH2	MSH3	MSH6	MTOR	MUTYH
MYCL (MYCL1)	MYCN	MYD88	MYO18A	NCOR2	NCSTN	NF1	NF2
NFKB1A	NKX2-1	NOD1	NOTCH1	NOTCH2	NPM1	NRAS	NT5C2
NTRK2	NTRK3	NUP93	NUP98	P2RY8	PAG1	PAK3	PALB2
PAX5	PBRM1	PC	PCBP1	PCL0	PDCD1	PDCD11	PDCD1LG2 (PD-L2)
PDGFRB	PDK1	PHF6	PIK3CA	PIK3CG	PIK3R1	PIK3R2	PIM1
POT1	PPP2R1A	PRDM1	PRKAR1A	PRKDC	PRSS8	PTCH1	PTEN
PTPN2	PTPN6 (SHP-1)	PTPRO	RAD21	RAD50	RAD51	RAF1	RARA
RB1	RELN	RET	RHOA	RICTOR	RNF43	ROS1	RPTOR
S1PR2	SDHA	SDHB	SDHC	SDHD	SERP2	SETBP1	SETD2
SGK1	SMAD2	SMAD4	SMARCA1	SMARCA4	SMARCB1	SMC1A	SMC3
SOC1	SOC2	SOC3	SOX10	SOX2	SPEN	SPOP	SRC
STAG2	STAT3	STAT4	STAT5A	STAT5B	STAT6	STK11	SUFU
TAF1	TBL1XR1	TCF3 (E2A)	TCL1A (TCL1)	TET2	TGFBR2	TLL2	TMEM30A
TMSB4XP8 (TMSL3)		TNFAIP3	TNFRSF11A	TNFRSF14	TNFRSF17	TOP1	TP53
TRAF2	TRAF3	TRAF5	TSC1	TSC2	TSHR	TUSC3	TYK2
U2AF2	VHL	WDR90	WHSC1 (MMSET or NSD2)		WISP3	WT1	XBP1
YYIAP1	ZMYM3	ZNF217	ZNF24 (ZSCAN3)	ZNF703	ZRSR2		

*Note: the assay was updated on 11/8/2016 to include the detection of alterations in CALR

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APPENDIX
Genes Assayed in FoundationOne®Heme
HEMATOLOGICAL MALIGNANCY DNA GENE LIST: FOR THE DETECTION OF SELECT REARRANGEMENTS

ALK	BCL2	BCL6	BCR	BRAF	CCND1	CRLF2	EGFR	EPOR
ETV1	ETV4	ETV5	ETV6	EWSR1	FGFR2	IGH	IGK	IGL
JAK1	JAK2	KMT2A (MLL)	MYC	NTRK1	PDGFRA	PDGFRB	RAF1	RARA
RET	ROS1	TMPRSS2	TRG					

HEMATOLOGICAL MALIGNANCY RNA GENE LIST: FOR THE DETECTION OF SELECT REARRANGEMENTS

ABI1	ABL1	ABL2	ACSL6	AFF1	AFF4	ALK	ARHGAP26 (GRAF)	
ARHGEF12	ARID1A	ARNT	ASXL1	ATF1	ATG5	ATIC	BCL10	BCL11A
BCL11B	BCL2	BCL3	BCL6	BCL7A	BCL9	BCOR	BCR	BIRC3
BRAF	BTG1	CAMTA1	CARS	CBFA2T3	CBBF	CBL	CCND1	CCND2
CCND3	CD274 (PD-L1)	CDK6	CDX2	CHIC2	CHN1	CIC	CIITA	CLP1
CLTC	CLTCL1	CNTRL (CEP110)	COL1A1	CREB3L1	CREB3L2	CREBBP	CRLF2	CSF1
CTNNB1	DDIT3	DDX10	DDX6	DEK	DUSP22	EGFR	EIF4A2	ELF4
ELL	ELN	EML4	EP300	EPOR	EPS15	ERBB2	ERG	ETS1
ETV1	ETV4	ETV5	ETV6	EWSR1	FCGR2B	FCRL4	FEV	FGFR1
FGFR10P	FGFR2	FGFR3	FLI1	FBNP1	FOXO1	FOXO3	FOXO4	FOXP1
FSTL3	FUS	GAS7	GLI1	GMPS	GPHN	HERPUD1	HEY1	HIP1
HIST1H4I	HLF	HMGA1	HMGA2	HOXA11	HOXA13	HOXA3	HOXA9	HOXC11
HOXC13	HOXD11	HOXD13	HSP90AA1	HSP90AB1	IGH	IGK	IGL	IKZF1
IL21R	IL3	IRF4	ITK	JAK1	JAK2	JAK3	JAZF1	KAT6A (MYST3)
KDSR	KIF5B	KMT2A (MLL)	LASP1	LCP1	LMO1	LMO2	LPP	LYL1
MAF	MAFB	MALT1	MDS2	MECOM	MKL1	MLF1	MLLT1 (ENL)	MLLT10 (AF10)
MLLT3	MLLT4 (AF6)	MLLT6	MN1	MNX1	MSI2	MSN	MUC1	MYB
MYC	MYH11	MYH9	NACA	NBEAP1 (BCL8)	NCOA2	NDRG1	NF1	NF2
NFKB2	NIN	NOTCH1	NPM1	NR4A3	NSD1	NTRK1	NTRK2	NTRK3
NUMA1	NUP214	NUP98	NUTM2A	OMD	P2RY8	PAFAH1B2	PAX3	PAX5
PAX7	PBX1	PCM1	PCSK7	PDCD1LG2 (PD-L2)	PDE4DIP	PDGFB	PDGFRA	PDGFRB
PER1	PHF1	PICALM	PIM1	PLAG1	PML	POU2AF1	PPP1CB	PRDM1
PRDM16	PRRX1	PSIP1	PTCH1	PTK7	RABEP1	RAF1	RALGDS	RAP1GDS1
RARA	RBM15	RET	RHOH	RNF213	ROS1	RPL22	RPN1	RUNX1
RUNX1T1 (ETO)	RUNX2	SEC31A	SEPT5	SEPT6	SEPT9	SET	SH3GL1	SLC1A2
SNX29 (RUNDC2A)	SRSF3	SS18	SSX1	SSX2	SSX4	STAT6	STL	SYK
TAF15	TAL1	TAL2	TBL1XR1	TCF3 (E2A)	TCL1A (TCL1)	TEC	TET1	TFE3
TFG	TFPT	TFRC	TLX1	TLX3	TMPRSS2	TNFRSF11A	TOP1	TP63
TPM3	TPM4	TRIM24	TRIP11	TTL	TYK2	USP6	WHSC1 (MMSET or NSD2)	
WHSC1L1	YPEL5	ZBTB16	ZMYM2	ZNF384	ZNF521			

ADDITIONAL ASSAYS: FOR THE DETECTION OF SELECT CANCER BIOMARKERS

Microsatellite (MS) status
Tumor Mutational Burden (TMB)

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APPENDIX
Performance Specifications

The median exon coverage for this sample is 879x

ACCURACY

Sensitivity: Base Substitutions	At $\geq 5\%$ Minor Allele Frequency	>99.0%
Sensitivity: Insertions/Deletions (1-40bp)	At $\geq 10\%$ Minor Allele Frequency	98.0%
Sensitivity: Focal Copy Number Alterations (Homozygous Deletions or Amplifications)	At $\geq 8\%$ copies	>95.0%
Sensitivity: Microsatellite status	At $\geq 20\%$ tumor nuclei	97.0%
Sensitivity: Known Gene Fusions	>95.0%	
Specificity: Base Substitutions, Insertions/Deletions, and Focal Copy Number Alterations	Positive Predictive Value (PPV)	>99.0%
Specificity: Known Gene Fusions	Positive Predictive Value (PPV)	>95.0%
Specificity: Microsatellite status	Positive Predictive Value (PPV)	>95.0%
Accuracy: Tumor Mutation Burden	At $\geq 20\%$ tumor nuclei	>90.0%
Reproducibility (average concordance between replicates)	97.0% inter-batch precision 97.0% intra-batch precision 95.0% microsatellite status precision 96.0% tumor mutation burden precision	

Assay specifications were determined for pical median exon coverage of approximately 500X. For additional information regarding the validation of FoundationOne, please refer to the article, Frampton, GM. et al. Development and validation of a clinical cancer genomic profiling test based on massively parallel DNA sequencing, Nat Biotechnol (2013 Oct. 20).

Microsatellite status, which is a measure of microsatellite instability (MSI), is determined by assessing indel characteristics at 114 homopolymer repeat loci in or near the targeted gene regions of the FoundationOne Heme test. Microsatellite status is assayed for all FoundationOne Heme samples and may be reported as "MSI-High", "MSI-Intermediate", "MS-Stable", or "Cannot Be Determined". Microsatellite status is reported as "Cannot Be Determined" if the sample is not of sufficient quality to be confidently determined.

Tumor Mutational Burden (TMB) is determined by measuring the number of somatic mutations occurring in sequenced genes on the FoundationOne and FoundationOne Heme tests and extrapolating to the genome as a whole. TMB is assayed for all FoundationOne and FoundationOne Heme samples and may be reported as "TMB-High", "TMB-Intermediate", "TMB-Low", or "Cannot Be Determined". TMB results, which are rounded to the nearest integer, are determined as follows: TMB-High corresponds to greater than or equal to 20 mutations per megabase (Muts/Mb); TMB-Intermediate corresponds to 6-19 Muts/Mb; TMB-Low corresponds to fewer than or equal to 5 Muts/Mb. Tumor Mutational Burden is reported as "Cannot Be Determined" if the sample is not of sufficient quality to confidently determine Tumor Mutational Burden.

For additional information specific to the performance of this specimen, please contact Foundation Medicine, Inc. at 1-888-988-3639.

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APPENDIX

About FoundationOne®Heme

ABOUT FOUNDATIONONE HEME

FoundationOne®Heme is a comprehensive genomic profiling test for hematologic malignancies and sarcomas. The test is designed to provide physicians with clinically actionable information to help with diagnostic sub-classification, prognosis assessment, and targeted therapeutic selection. Test results provide information about clinically significant alterations, potential targeted therapies, available clinical trials, and quantitative markers that may support immunotherapy clinical trial enrollment. FoundationOne Heme is analytically validated to detect all classes of genomic alterations in more than 400 cancer-related genes. In addition to DNA sequencing, FoundationOne Heme employs RNA sequencing across more than 250 genes to capture a broad range of gene fusions, common drivers of hematologic malignancies and sarcomas.

FoundationOne Heme was developed and its performance characteristics determined by Foundation Medicine, Inc. (Foundation Medicine). FoundationOne Heme has not been cleared or approved by the United States Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. FoundationOne Heme may be used for clinical purposes and should not be regarded as purely investigational or for research only. Foundation Medicine's clinical reference laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high-complexity clinical testing.

THE REPORT

Incorporates analyses of peer-reviewed studies and other publicly available information identified by Foundation Medicine; these analyses and information may include associations between a molecular alteration (or lack of alteration) and one or more drugs with potential clinical benefit (or potential lack of clinical benefit), including drug candidates that are being studied in clinical research. Note: A finding of biomarker alteration does not necessarily indicate pharmacologic effectiveness (or lack thereof) of any drug or treatment regimen; a finding of no biomarker alteration does not necessarily indicate lack of pharmacologic effectiveness (or effectiveness) of any drug or treatment regimen.

Diagnostic Significance

FoundationOne Heme identifies alterations to select cancer-associated genes or portions of genes (biomarkers). In some cases, the Report also highlights selected negative test results regarding biomarkers of clinical significance.

Qualified Alteration Calls (Equivocal and Subclonal)

An alteration denoted as "amplification – equivocal" implies that FoundationOne Heme data provide some, but not unambiguous, evidence that the copy

number of a gene exceeds the threshold for identifying copy number amplification. The threshold used in FoundationOne Heme for identifying a copy number amplification is five (5) for *ERBB2* and six (6) for all other genes. Conversely, an alteration denoted as "loss – equivocal" implies that FoundationOne Heme data provide some, but not unambiguous, evidence for homozygous deletion of the gene in question. An alteration denoted as "subclonal" is one that FoundationOne Heme analytical methodology has identified as being present in <10% of the assayed tumor DNA.

Ranking of Alterations and Therapies
Biomarker Findings

Appear at the top of the report, but are not ranked higher than Genomic Findings.

Genomic Findings

Therapies with Clinical Benefit In Patient's Tumor Type → Therapies with Clinical Benefit In Other Tumor Type → Clinical Trial Options → No Known Options (If multiple findings exist within any of these categories, the results are listed alphabetically by gene name.)

Therapies

Sensitizing therapies → Resistant therapies. (If multiple therapies exist within any of these categories, they are listed alphabetically by therapy name.)

Clinical Trials

Pediatric trial qualification → Geographical proximity → Later trial phase.

LEVEL OF EVIDENCE NOT PROVIDED

Drugs with potential clinical benefit (or potential lack of clinical benefit) are not evaluated for source or level of published evidence.

NO GUARANTEE OF CLINICAL BENEFIT

This Report makes no promises or guarantees that a particular drug will be effective in the treatment of disease in any patient. This Report also makes no promises or guarantees that a drug with potential lack of clinical benefit will in fact provide no clinical benefit.

NO GUARANTEE OF REIMBURSEMENT

Foundation Medicine makes no promises or guarantees that a healthcare provider, insurer or other third party payor, whether private or governmental, will reimburse a patient for the cost of FoundationOne Heme.

TREATMENT DECISIONS ARE RESPONSIBILITY OF PHYSICIAN

Drugs referenced in this Report may not be suitable for a particular patient. The selection of any, all or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) resides

entirely within the discretion of the treating physician. Indeed, the information in this Report must be considered in conjunction with all other relevant information regarding a particular patient, before the patient's treating physician recommends a course of treatment. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care in a given community. A treating physician's decisions should not be based on a single test, such as this Test, or the information contained in this Report. Certain sample or variant characteristics may result in reduced sensitivity. These include: subclonal alterations in heterogeneous samples, low sample quality or with homozygous losses of <3 exons; and deletions and insertions >40bp, or in repetitive/high homology sequences. FoundationOne Heme is performed using DNA and RNA derived from tumor, and as such germline events may not be reported.

The following targets typically have low coverage resulting in a reduction in sensitivity: SDHD exon 4, TNFRSF11A exon1, and TP53 exon 1.

FoundationOne Heme complies with all European Union (EU) requirements of the IVD Directive 98/79EC. As such, the FoundationOne Heme Assay has been registered for CE mark by our EU Authorized Representative, Qarad b.v.b.a, Pas 257, B-2440 Geel, Belgium.



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APPENDIX

About FoundationOne®Heme

SELECT ABBREVIATIONS

ABBREVIATION	DEFINITION
CR	Complete response
DCR	Disease control rate
DNMT	DNA methyltransferase
HR	Hazard ratio
ITD	Internal tandem duplication
MMR	Mismatch repair
mut/Mb	Mutations per megabase
NOS	Not otherwise specified
ORR	Objective response rate
OS	Overall survival
PD	Progressive disease
PFS	Progression-free survival
PR	Partial response
SD	Stable disease
TKI	Tyrosine kinase inhibitor

SAMPLE

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APPENDIX

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APPENDIX

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